MUSIIWA THANDABANTU

DATA ANALYTICS

Exercise 3: SQL CASE Statement

1. Table: products

product_id	product_name	price
1	Laptop	1200.00
2	phone	800.00
3	keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

Classify each product by price:

* 'Expensive' if price > 1000
* 'mid-range' if price between 100 and 1000
* 'Budget' if price <100

```
SELECT product_id, product_name, price,
CASE
WHEN price >1000 THEN 'Expensive'
WHEN price BETWEEN 100 AND 1000 THEN 'Mid-range'
WHEN price <100 THEN 'Budget'
END AS price_category
FROM products;
```

product_id	product_name	price	price_category
1	Laptop	1200.00	Expensive
2	phone	800.00	Mid-range
3	Keyboard	45.00	Budget
4	Monitor	300.00	Mid-range
5	Mouse	25.00	Budget

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② Table : orders

order_id	Customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Label each order :

* 'High value' for orders ≥ 1000
* 'medium value' for 500 - 999.99
* 'Low value' for orders < 500

```
SELECT order_id, customer_name, amount,
CASE
WHEN orders ≥ 1000 THEN 'High value'
WHEN orders BETWEEN 500 - 999.99 THEN 'Medium value'
WHEN orders < 500 THEN 'Low value'
END AS order_value_category
FROM orders;
```

order_id	customer_name	amount	order_value_category
1	ALICE	150.00	Low value
2	Bob	560.00	Medium value
3	Charlie	999.99	Medium value
4	Diana	45.50	Low value
5	Ethan	1200.00	High value

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③ Table : employees

emp_id	emp_name	department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

- **Categorize employee position :**

* If in 'IT' and Salary > 80000 -> 'Senior IT'
* If in 'HR' and Salary > 55000 -> 'Experienced HR'
* Otherwise -> 'Staff'

```
SELECT emp_id, emp_name, department, Salary,
CASE
WHEN salary > 80000 THEN 'Senior IT'
WHEN Salary > 55000 THEN 'Experienced HR'
ELSE Staff.
END AS position_level
FROM employees;
```

emp_id	emp_name	department	Salary	position_level
1	John	IT	85000	Senior IT
2	Sara	HR	60000	Experienced HR
3	Mark	IT	75000	Senior IT
4	Lucy	Finance	95000	Staff
5	Tom	HR	55000	NULL

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④ Table: Students

Student_id	Student_name	Score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

Assign a letter grade:

* ≥ 90: 'A'
* 80-89: 'B'
* 70-79: 'C'
* 60-69: 'D'
* < 60: 'F'

```
SELECT student_name, score,
CASE
WHEN grade ≥ 90 THEN 'A'
WHEN grade BETWEEN 80 AND 89 THEN 'B'
WHEN grade BETWEEN 70 AND 79 THEN 'C'
WHEN grade BETWEEN 60 AND 69 THEN 'D'
WHEN grade < 60 THEN 'F'
END AS grade
FROM Students;
```

Student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

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⑤ Table : deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Label delivery performance :

* ≤ 30 mins : 'Fast'
* 31 – 60 mins : 'On Time'
* 60 mins : 'Late'

```
SELECT delivery_id, delivery_time_minutes,
CASE
WHEN mins ≤ 30 THEN 'Fast'
WHEN mins BETWEEN 31 AND 60 THEN 'On Time'
WHEN mins = 60 THEN 'Late'.
END AS performance
FROM deliveries;
```

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	NULL
3	30	FAST
4	65	NULL
5	100	NULL

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(6) Table: tickets

tickets_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	password reset	1

Convert priority to labels

- 3 -> 'High'
- 2 -> 'Medium'
- 1 -> 'Low'

```
SELECT issue_type, priority;
CASE
WHEN priority_column = 3 THEN 'High'
WHEN priority_column = 2 THEN 'Medium'
WHEN priority_column = 1 THEN 'Low'
END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
password reset	1	Low

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⑦ Table: attendance

Student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Calculate attendance % and classify:

* ≥ 90% → 'Excellent'
* 75-89% → 'Good'
* <75% → 'Needs improvement'

```
SELECT student_id, attendance_percentage
CASE
WHEN percentage ≥ 90 THEN 'Excellent'
WHEN percentage BETWEEN 75 AND 89 THEN 'GOOD'
WHEN percentage <75 THEN 'Needs improvement'
END AS attendance_status
FROM attendance;
```

Student_id	attendance percentage,	attendance_status
1	45	Excellent
2	30	Needs improvement
3	48	Excellent
4	25	Needs improvement
5	50	Excellent

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② Table: product_inventory

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Label stock status:

* 0 -> 'out of stock'
* 1-5 -> 'Low stock'
* 5 -> 'in stock'

```
SELECT product_id, stock_qty,
CASE
WHEN stock = 0 THEN 'out of stock'
WHEN stock BETWEEN 1 AND 5 THEN 'Low stock'
WHEN stock > 5 THEN 'in stock'
END AS stock_status
FROM product_inventory;
```

product_id	stock_qty	stock_status
1	5	Low stock
2	0	out of stock
3	25	in stock
4	10	in stock
5	3	Low stock

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⑨ Table: Classes

Class_id	Subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

Classify by size:

* ≥ 25 -> 'Large'
* 10 - 24 -> 'Medium'
* < 10 -> 'Small'

```
SELECT subject, enrolled_students,
CASE
WHEN size ≥ 25 THEN 'Large'
WHEN size BETWEEN 10 AND 24 THEN 'Medium'
WHEN size < 10 THEN 'Small'
END AS class_size_category
FROM classes;
```

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

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(10) Table: payments

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	card
4	75.00	paypal
5	300.00	Cash

Apply discount flag:

* If payment_method = 'cash and amount ≥ 200' 'Eligible for discount'
* otherwise → 'NOT Eligible'

```
SELECT payment_id, payment_method, amount,
CASE
WHEN ~~discount~~ payment_method = 'cash' AND
amount ≥ 200 THEN 'Eligible for discount'
ELSE 'Not Eligible'
END AS discount_flag
FROM payments;
```

payment_id	payment_method	amount	discount_flag
1	card	50.00	NOT Eligible
2	cash	200.00	Eligible for discount
3	Card	150.00	Not Eligible
4	paypal	75.00	Not Eligible
5	cash	300.00	Eligible for discount

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